FINAL PROJECT REPORT
Pneumonia Detection using Computer Vision

Submitted by: Rajasekar T
Date: 31-May-2026

Table of Contents

1. Executive Summary

This project focuses on building a computer vision model to detect pneumonia from chest X-ray images. The solution leverages deep learning techniques, specifically Convolutional Neural Networks (CNN), to identify lung opacity patterns associated with pneumonia. The objective is to assist medical professionals in faster and more accurate diagnosis, reducing dependency on manual interpretation and enabling scalable healthcare solutions.

2. Problem Statement

Pneumonia diagnosis relies heavily on radiologists interpreting chest X-rays. This creates bottlenecks in high-demand environments and increases the risk of diagnostic errors. The goal of this project is to develop an automated system that classifies X-ray images into pneumonia and normal categories, thereby supporting clinical decision-making.

3. Data Overview

The dataset consists of approximately 30,000 chest X-ray images labeled based on the presence of lung opacity. Images were processed from DICOM format and converted into arrays for model training and faster processing. The dataset is structured for binary classification.

Observations:

- Some images are darker / low contrast
- Some contain artifacts (text overlays, markers, borders)
- Lung regions are central; background noise exists
- The image sizes are not uniform and varies across
- Images are grayscale medical X-rays
- Pixel intensity varies significantly across images
- Data comes from multiple machines / environments
- Model must generalize across variations

4. Exploratory Data Analysis

EDA involved visualizing sample images and analyzing class distribution. Pneumonia cases show visible opacity in lung regions, while normal cases show clear structures. Potential class imbalance was identified, which may impact model performance.
Class imbalance observed:

- Class 0 (No Lung Opacity) ≈ 65–70%
- Class 1 (Lung Opacity) ≈ 30–35%

Binary labelling:
- 1 = Lung opacity (possible pneumonia)
- 0 = Normal OR abnormal but not opacity

Important nuance:
- Class 0 includes both normal and other abnormalities
- Model is detecting "opacity", not full diagnosis

Pneumonia (opacity) appears as:
- White / hazy regions in lungs
- Often diffused, not sharply bounded

No overlap between train and test sets, which ensures:
- No data leakage
- Reliable evaluation

Exact overlap between class file and train file

Figure 1: Sample X-ray Images (Label = 0 indicate no Normal/ Not Normal')

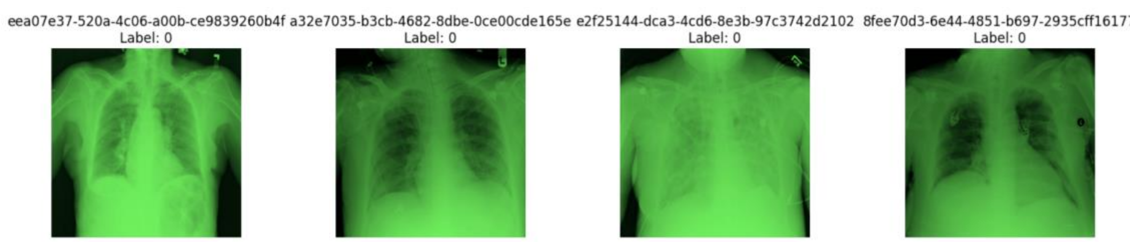


Figure 2: Sample X-ray Images (Label = 1 indicate Pneumonia)

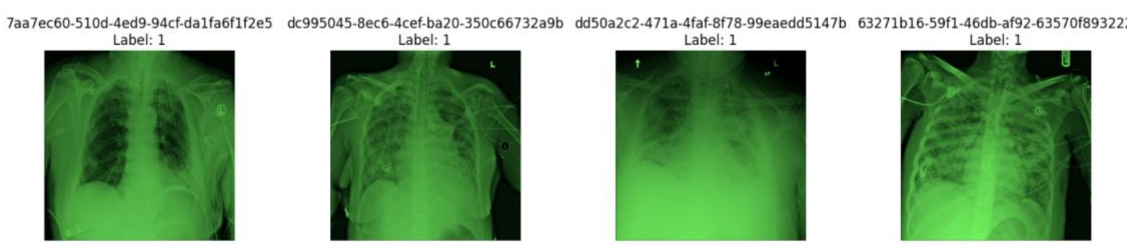


Figure 3: Sample X-ray Images (Class=Normal)

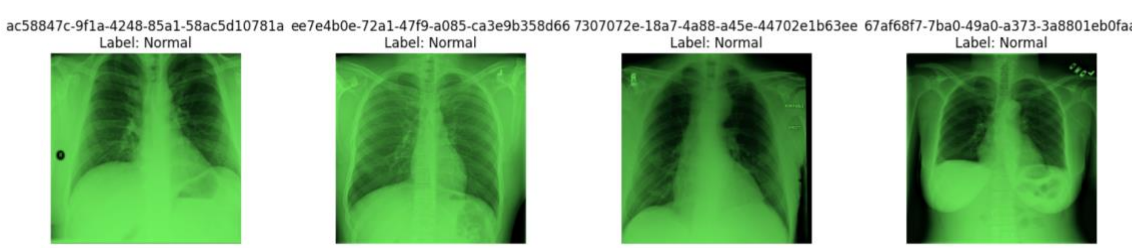
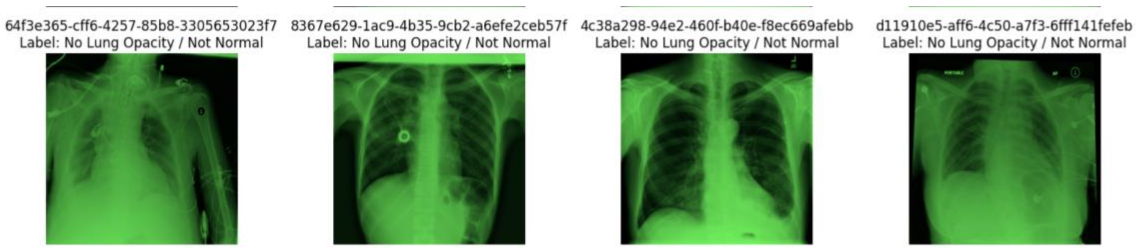


Figure 4: Sample X-ray Images (Class= 'No Lung Opacity / Not Normal')



Key Insight

- From the above diagrams it is very clear that, there are 'No Lung Opacity / Not Normal' (from Figure 4) cases that are Labelled as 0
- This might confuse the model and there could be a lot of False positives

5. Data Preprocessing

Images are provided in zip format, these are converted into raw array format using a pipeline and then converted into the processed array. In this way, the repeated task of converting the images (potential bottleneck) can be minimized in the repeated iteration and the model execution will be faster.

Image processing:

- Since the provided images are already in grayscale, there is no conversion done.
- Images were normalized so that the brightness of the images fits perfectly inside a new range, like 0 to 255.
- The Images are resized into two aspects 256x256 and 224x224 for better focus and model performance. This will help in removing unwanted details or noise on the images and helps model focus on the opacity of the lungs.

Class weightage:

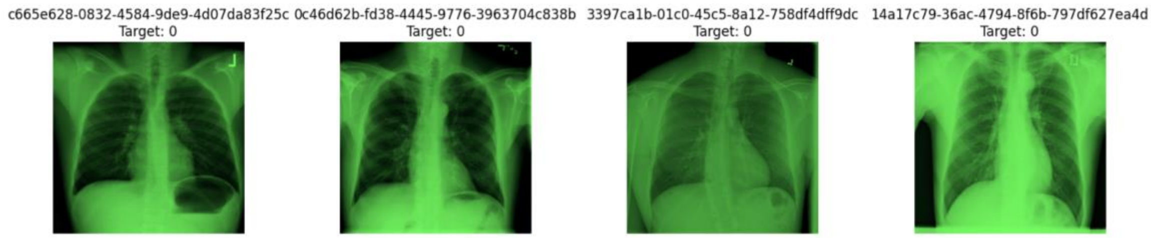
- Because of the imbalance in the data given, the model is performed on a slightly higher weightage for the positive cases.

Train and Validation:

- Since there is already a test data given without any labels.
- The training dataset was split into training and validation. These steps improve computational efficiency and model stability.
- Post model training, the same model will be used for predicting the test data.

Figure 5: Before Pre-processing

Figure 6: After Pre-processing



Key Insight

- From the above image comparison, it is evident that the images that are processed looks crisp and clear when compared to the original images.

6. CNN Model Building

A Convolutional Neural Network (CNN) was developed using a combination of convolutional, pooling, and fully connected layers. The model was trained on the pre-processed dataset and evaluated using validation data to assess its performance and generalization.

Given the critical nature of medical diagnosis, the model is intentionally designed to **prioritize the detection of positive (pneumonia) cases**. Specifically, higher importance is assigned to minimizing **False Negatives**, as failing to identify a pneumonia case could prevent patients from receiving timely medical attention, potentially leading to severe or fatal outcomes.

To address this, the model incorporates class weighting to give higher emphasis to positive cases during training. While standard metrics such as accuracy, precision, and recall are evaluated, greater focus is placed on **recall and sensitivity**, as they directly reflect the model's ability to correctly identify pneumonia cases.

Additionally, evaluation metrics such as **Precision-Recall (PR) Curve and ROC-AUC** on validation data are used to further assess the model's effectiveness in capturing positive cases and ensuring minimal miss-outs. These metrics provide deeper insights into the trade-offs between precision and recall, which is crucial in a high-risk medical context.

The model training started with a threshold of 0.5 later it was tuned to 0.33 to 0.38 which significantly improved the F1 and precision.

7. Model Evaluation:

As highlighted earlier, the model training and evaluation happened in 2 different image size

7.1 Model Performance Summary (256 × 256)

(Ref: Figure 7, Figure 8, and Figure 9)

The model shows steady improvement across epochs, with training and validation metrics closely aligned, indicating good generalization and no significant overfitting.

PR-AUC trends confirm improved capability in identifying positive (pneumonia) cases, which is critical given the class imbalance.

At the optimized threshold (~0.38), the model achieves:

- **~75% recall for pneumonia cases**
- **~71% overall accuracy**

This reflects a deliberate bias toward minimizing **False Negatives**, ensuring high-risk patients are not missed.

While precision (~53%) indicates some false positives, this is an acceptable trade-off in a medical context where missed diagnoses carry significantly higher risk.

7.2 Model Performance Summary (224 × 224)

(Ref: Figure 10, Figure 11, Figure 12)

The model demonstrates steady learning with aligned training and validation trends, indicating good generalization.

PR-AUC shows consistent improvement, confirming enhanced capability in detecting pneumonia cases.

At the optimized threshold:

- **Recall (Pneumonia): ~75%**
- **Accuracy: ~71–72%**

The model is intentionally tuned to **minimize False Negatives**, ensuring critical cases are not missed, even at the cost of moderate false positives.

7.3 Comparison

(Refer Appendix Table1 and Table2)

Metric	224 × 224	256 × 256
Accuracy	~0.71	~0.72
Recall (Pneumonia)	~0.75	~0.77
Precision	~0.53	~0.54
F1 Score	~0.62	~0.64
ROC-AUC	~0.77	~0.79
PR-AUC	~0.57–0.58	~0.61–0.62

Key Insight

- The 256 × 256 model shows marginal but consistent improvement across all metrics, especially in recall and PR-AUC, making it more suitable for medical use where missing positive cases is critical.

8. Test data prediction

The model after training, it is used to predict the Pneumonia on the test dataset and following are the result.

Prediction	224 × 224	256 × 256
Negative (0)	59.73%	59.67%
Positive (1)	40.27%	40.33%

Key Insight:

- Both models produce **nearly identical prediction distributions**, indicating stable model behavior across input sizes.
- The 256 × 256 model shows **slightly higher positive detection**, aligning with its improved recall and PR-AUC.
- While prediction distribution remains consistent, the **256 × 256 model delivers better detection performance**, making it the preferred choice for minimizing missed pneumonia cases.

9. Key Insights & Business Implications

The CNN-based model demonstrates **stable learning and strong generalization**, effectively capturing pneumonia-related patterns from chest X-ray images. Performance trends indicate that the model is reliable and consistent across training and validation datasets.

A key strength of the model is its **high recall (~75–77%)**, ensuring that the majority of pneumonia cases are correctly identified. This is critical in healthcare, where missing a positive case (False Negative) can lead to severe clinical consequences. The model is therefore intentionally optimized to **prioritize sensitivity over precision**.

From a business perspective, the model is well-suited as a **screening and triage tool**, particularly in:

- Resource-constrained environments
- High-volume diagnostic settings

While the model generates some false positives, this is an acceptable trade-off, as it ensures that high-risk patients are not overlooked. Additionally, **threshold tuning and resolution optimization (224 vs 256)** provide flexibility to balance performance and computational efficiency based on deployment needs.

Overall, the solution serves as a **robust baseline AI system** that can augment clinical workflows, reduce radiologist workload, and improve early detection rates.

10. Transfer Learning Solution:

There are many transfer learning methodology available which are pre-trained on multiple images. The famous transfer learning models for this use-case are **DenseNet121** and **EfficientNet**

- **DenseNet121** is widely adopted in medical imaging applications due to its ability to reuse features across layers, enabling better detection of subtle abnormalities in X-ray images while reducing the risk of vanishing gradients.

- **EfficientNet** provides an optimized balance between model size and performance by scaling network depth, width, and resolution simultaneously. This often results in higher accuracy with lower computational cost.

We will try with DenseNet121 as our primary goal is to get highest recall and balancing the accuracy.

10.1 Why DenseNet121?

- DenseNet121 was selected because its dense connectivity architecture enables efficient feature reuse and stronger feature extraction, making it highly effective for medical image classification tasks.
- Compared to the CNN models developed from scratch, DenseNet121 achieved significantly higher Accuracy, Recall, ROC-AUC, and PR-AUC, demonstrating superior capability in identifying pneumonia-related patterns in chest X-ray images.

11. DenseNet121 - Base Model:

- Without modifying or fine tuning let's try training and evaluating the performance of the DenseNet121 base model when compared to our CNN approach
- As part of the pre-processing, we keep the image size of 244x244 as it was throwing the highest recall in our previous CNN approach and it will help the model to be more focused
- Also, to execute the pipeline faster on all the given images of 30,000. Lets convert the images into modules of 2,000 for faster training.

11.1 Model Performance Summary:

(Ref: Figure 13)

- DenseNet121 significantly outperformed the CNN (224×224) model, improving **Accuracy** from **72.4% to 78.3%**, **Recall** from **76% to 79%,** and **PR-AUC** from **0.62 to 0.74**
- The improvement demonstrates that transfer learning and pre-trained feature extraction are substantially more effective than training a CNN from scratch for detecting pneumonia patterns in chest X-ray images.
- The best threshold which came out with best performing metrics was **0.38**

Accuracy	Recall	Precision	F1	ROC-AUC	PR-AUC
0.782997	0.793512	0.623117	0.698067	0.861636	0.738858

12. DenseNet121 – Fine Tuned Model:

- DenseNet121 was initially trained using pre-trained ImageNet weights with the base network frozen, allowing only the custom classification layers to learn pneumonia-specific patterns.

- Fine-tuning was then performed by unfreezing the last few DenseNet layers and retraining them with a very low learning rate (1e-5), enabling the model to adapt high-level image features to the chest X-ray domain while preserving the learned representations from the pre-trained model.

12.1 Model Performance Summary:

(Ref: Figure 14)

- Fine-tuning DenseNet121 improved Recall from **79% to 80.8%** and reduced false negatives, enabling the model to identify more pneumonia cases.
- While Accuracy and Precision remained largely stable, the fine-tuned model is better suited for medical screening scenarios where maximizing case detection is a higher priority than minimizing false positives.
- The best threshold which came out with best performing metrics was **0.31**

Accuracy	Recall	Precision	F1	ROC-AUC	PR-AUC
0.780792	**0.808162**	**0.617044**	**0.699789**	**0.862862**	**0.741479**

- Also, we have tried with the various thresholds between 0.3 to 0.4 to check whether the recall can be increased further without compromising on the accuracy.

13. Model Comparison and Selection:

- Now, lets compare the model performance that we have tried.

Model	Threshold	Accuracy	Recall	Precision	F1 Score	ROC-AUC	PR-AUC
CNN (256x256)	0.380	0.7141	0.7496	0.5340	0.6237	0.7788	0.5780
CNN (224x224)	0.342	0.7236	0.7691	0.5444	0.6376	0.7975	0.6196
DenseNet121 (Frozen)	0.353	0.7858	0.7851	0.6291	0.6985	0.8627	0.7424
DenseNet121 (Fine-Tuned Best)	0.313	0.7808	0.8082	0.6170	0.6998	0.8629	0.7415
DenseNet121 (FT @ 0.32)	0.320	0.7818	0.8008	0.6199	0.6988	0.8629	0.7415
DenseNet121 (FT @ 0.38)	0.380	0.7948	0.7412	0.6551	0.6955	0.8629	0.7415

13.1 Key Insights from Model Comparison

- Reducing image size from 256×256 to 224×224 improved model performance while reducing computational cost.

- Transfer learning using DenseNet121 significantly outperformed CNN models trained from scratch across all evaluation metrics.
- DenseNet121 achieved approximately 8–10% improvement in accuracy and 20% improvement in PR-AUC compared to the baseline CNN.
- Fine-tuning the DenseNet model increased Recall from ~78.5% to ~80.8%, reducing the number of missed pneumonia cases.
- Threshold optimization had a significant impact on model performance, highlighting the importance of selecting thresholds based on business objectives.
- Higher thresholds improved precision and accuracy, while lower thresholds improved recall.
- Performance gains were primarily driven by model architecture and transfer learning, rather than increasing image resolution.

13.2 Recommended Model

For Medical Screening Use Cases

DenseNet121 Fine-Tuned with Best Threshold (0.31) is the recommended model.

Achieved:
* Accuracy: 78.1%
* Recall: 80.8% (Highest)
* Precision: 61.7%
* F1 Score: 0.70
* ROC-AUC: 0.863
* PR-AUC: 0.741

This model minimizes False Negatives, ensuring that the maximum number of pneumonia cases are identified.
In healthcare screening scenarios, missing a pneumonia case is generally more critical than generating additional false positives, making this model the most suitable choice.

13.3 For Balanced Production Deployment

DenseNet121 Frozen can be considered due to its slightly higher precision and overall balance between recall and precision.

13.4 Final Takeaway

- Transfer learning proved to be the most impactful improvement in the project.
- Fine-Tuned DenseNet121 with the optimized threshold provides the best screening-oriented performance and is the preferred model when patient safety and early pneumonia detection are the primary objectives.

14. APP for Business

(Ref - Figure 15)

- Also, I have built a prototype app for the Hospital staff on Hugging face. They can upload the X-ray images and get the results and proceed with the further steps for each patients.
- Details:
    - https://huggingface.co/spaces/Rajse/pneumonia-detection

- This APP was built on top of the Fine-tuned DenseNet Model with Optimized Threshold

15. Limitations & Next Steps

15.1 Limitations

- The dataset contains a moderate class imbalance (~70% negative vs ~30% positive), which can influence model learning and prediction bias.
- The model is trained on a single chest X-ray dataset and may not fully generalize to images from different hospitals, imaging devices, or patient demographics.
- The current solution performs binary classification and does not localize or highlight the pneumonia region within the X-ray.
- Model predictions should be treated as a screening aid and not as a standalone clinical diagnosis.
- Resource limitations prevented extensive experimentation with larger transfer learning architectures and ensemble approaches.

15.2 Next Steps

- Implement explainable AI techniques such as Grad-CAM to visualize lung regions influencing the model's prediction.
- Evaluate additional transfer learning architectures such as EfficientNet and ConvNeXt for potential performance improvements.
- Experiment with advanced loss functions such as Focal Loss to further improve recall and reduce false negatives.
- Incorporate data augmentation and external datasets to improve model robustness and generalization.
- Develop a production-ready inference pipeline with model monitoring and automated retraining capabilities.
- Extend the solution to support pneumonia localization and severity assessment.

15.3 Business Recommendation

- Deploy the **Fine-Tuned DenseNet121 model** as the primary screening model due to its highest recall (~81%), minimizing missed pneumonia cases.
- Use the solution as a decision-support tool to prioritize high-risk cases for radiologist review, reducing diagnostic turnaround time.
- Implement threshold tuning based on operational requirements, balancing recall and precision for different healthcare workflows.
- Continuously retrain the model using newly collected X-ray images to maintain performance and reduce model drift.
- Introduce explainability features to improve clinician trust and support adoption within healthcare environments.

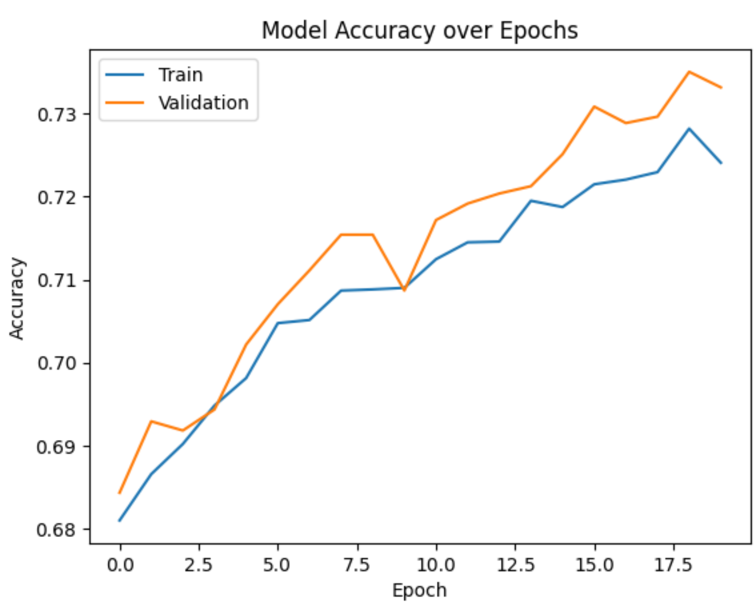
16. Final Recommendation:

- The **Fine-Tuned DenseNet121 model with an optimized threshold (~0.31)** is recommended for deployment in screening-oriented healthcare settings, as it provides the highest pneumonia detection rate while maintaining strong overall predictive performance.

17. Appendix

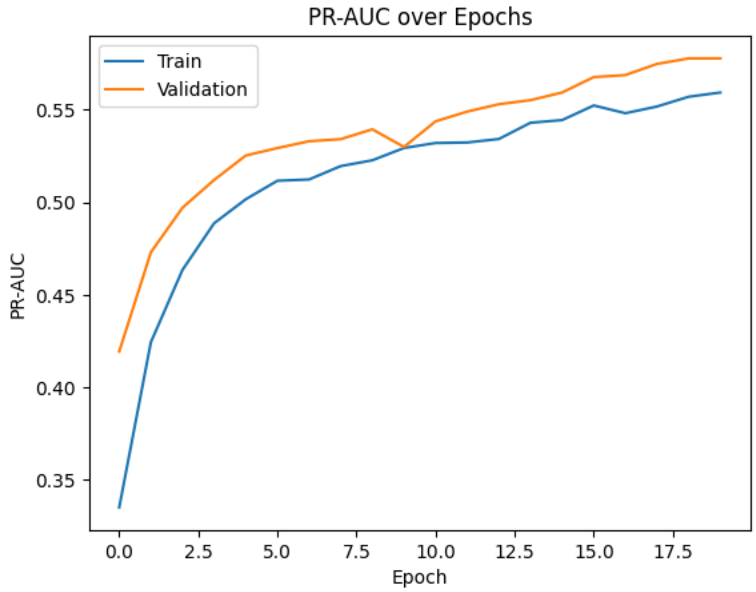
This section includes raw code, detailed outputs, and additional experiments.

Figure 7: Training Vs Validation – Model Accuracy (Image size – 256 x 256)



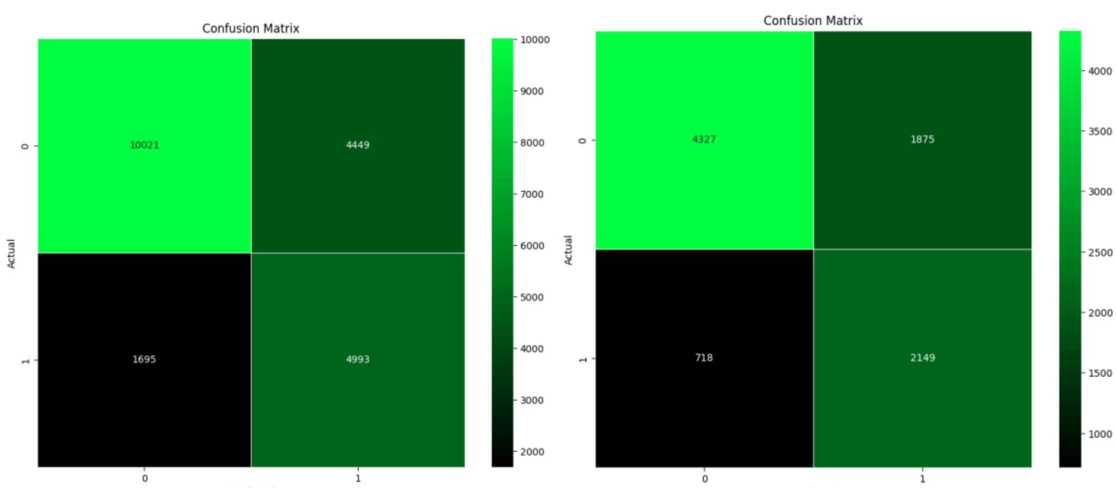
Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 8: Training Vs Validation – PR-AUC trend (Image size – 256 x 256)



Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 9: Training Vs Validation Confusion Matrix (Image size – 256 x 256)



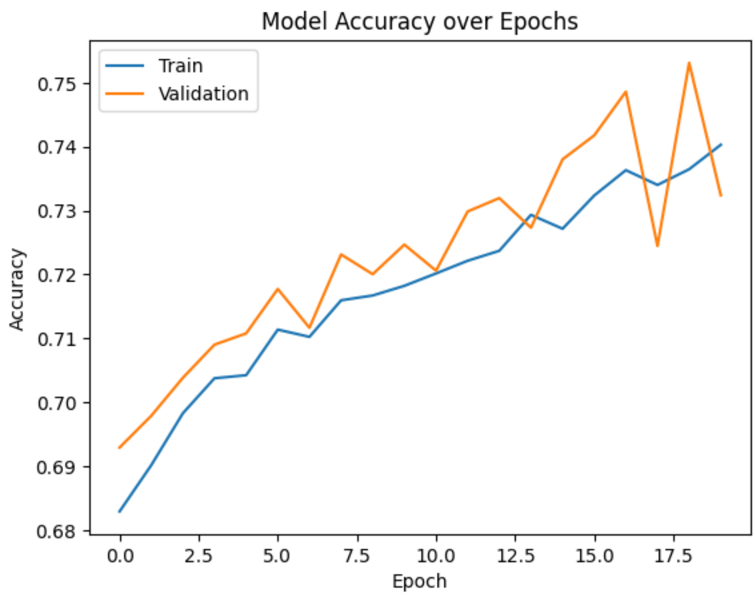
Training Validation

Table1: image size 256 x 256

	Accuracy	Recall (Class 1)	Precision (Class 1)	F1 (Class 1)	ROC-AUC	PR-AUC
Training	71%	75%	53%	62%	78%	57%

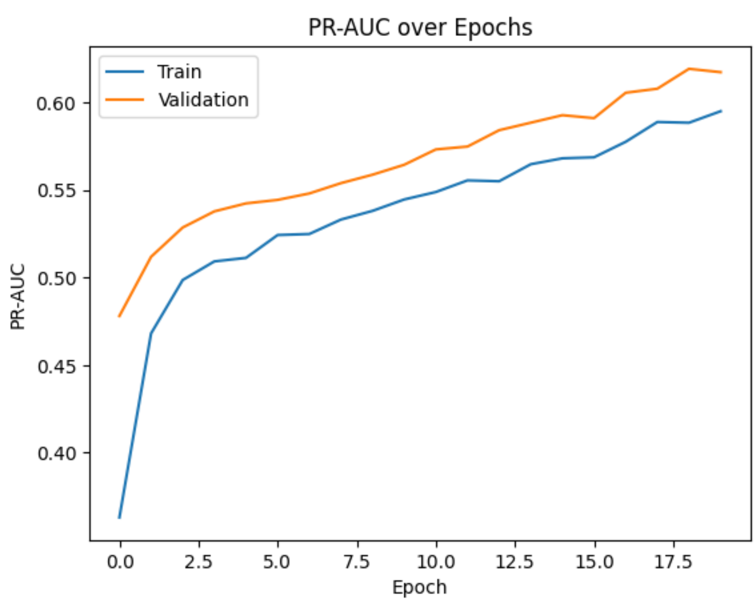
Validation	71%	75%	53%	62%	78%	58%

Figure 10: Training Vs Validation – Model Accuracy (Image size – 224 x 224)



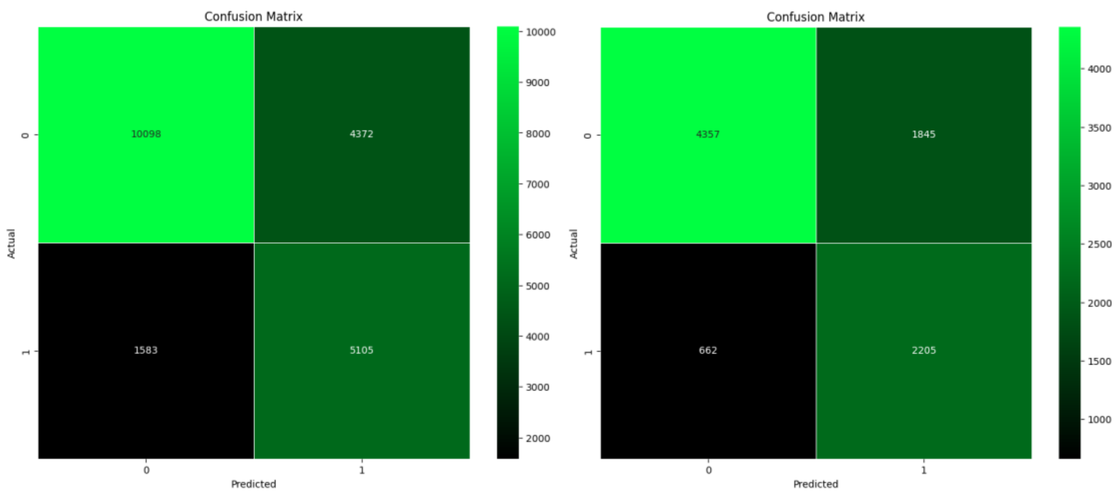
Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 11: Training Vs Validation – PR-AUC trend (Image size – 224 x 224)



Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 12: Training Vs Validation Confusion Matrix (Image size – 224 x 224)



Training Validation

Table 2: image size 224 x 224

	Accuracy	Recall (Class 1)	Precision (Class 1)	F1 (Class 1)	ROC-AUC	PR-AUC
Training	72%	76%	54%	63%	79%	61%
Validation	72%	77%	54%	64%	80%	62%

Figure 13: DenseNet121 Basemodel Training Vs Validation Confusion Matrix

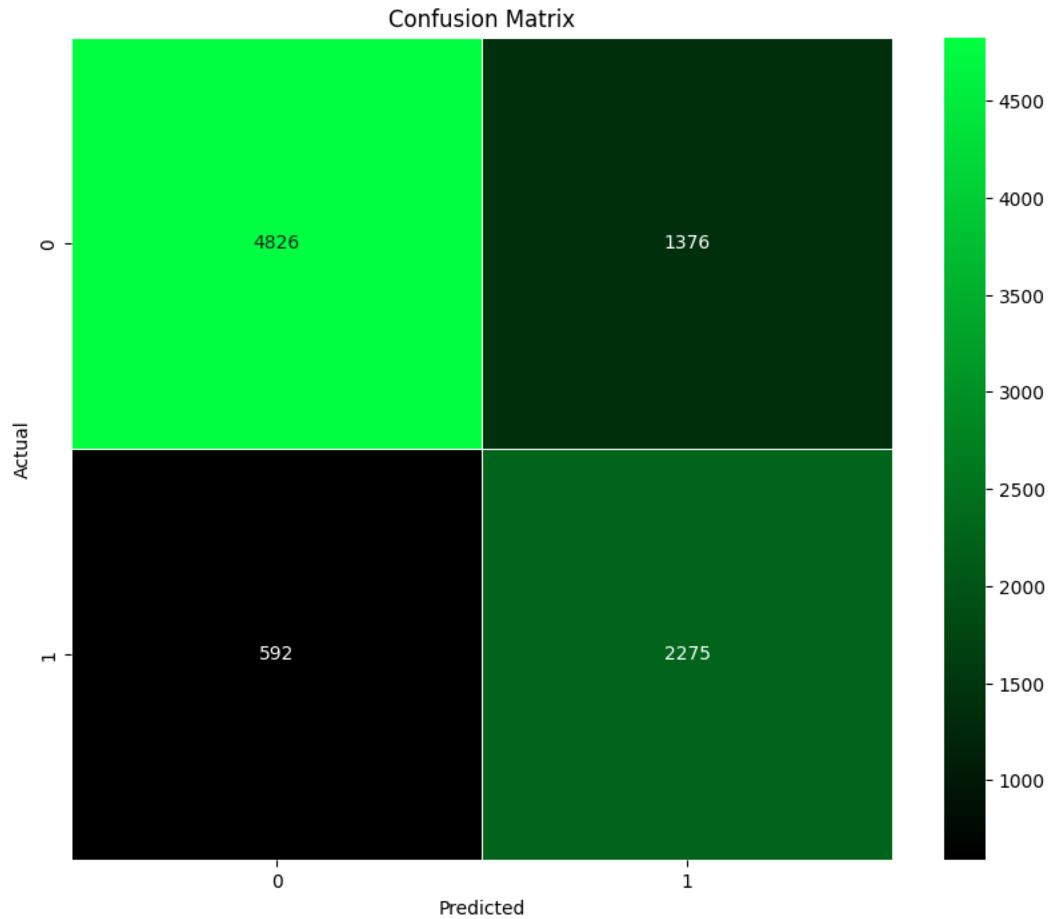


Figure 14: DenseNet121 Fine Tuned -- Training Vs Validation Confusion Matrix

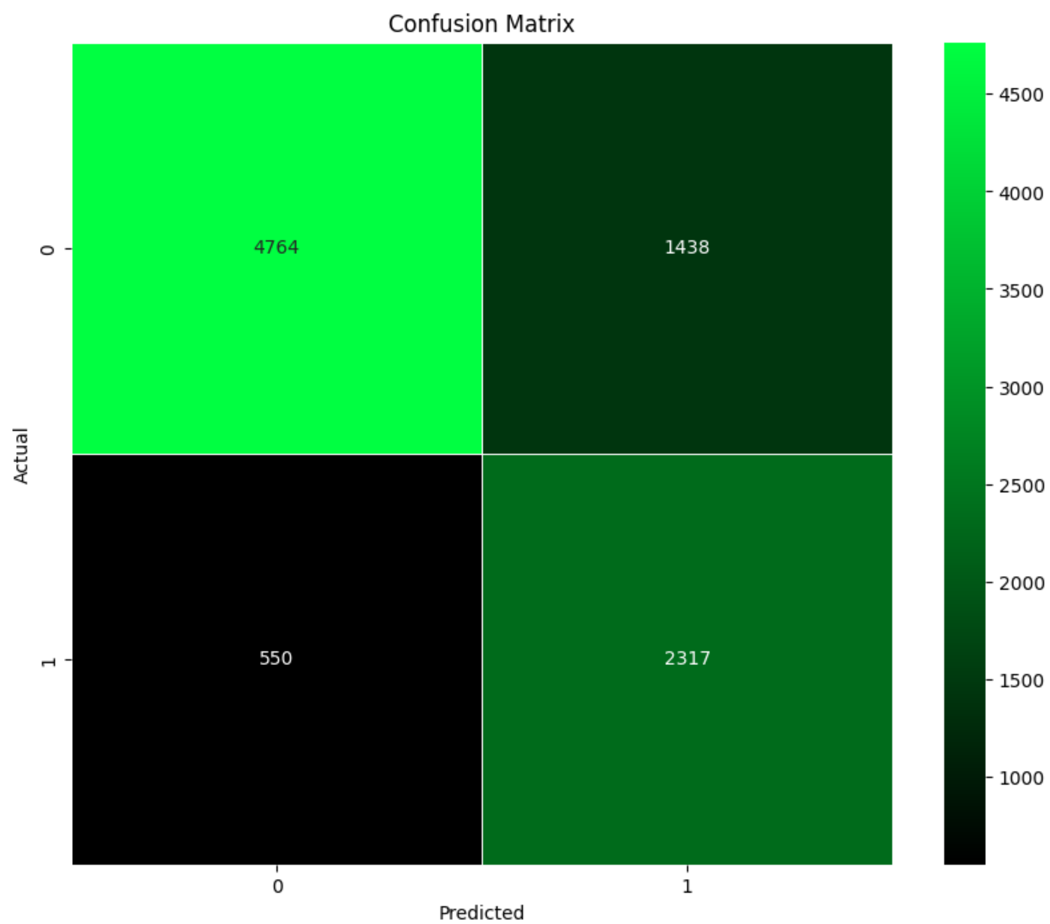
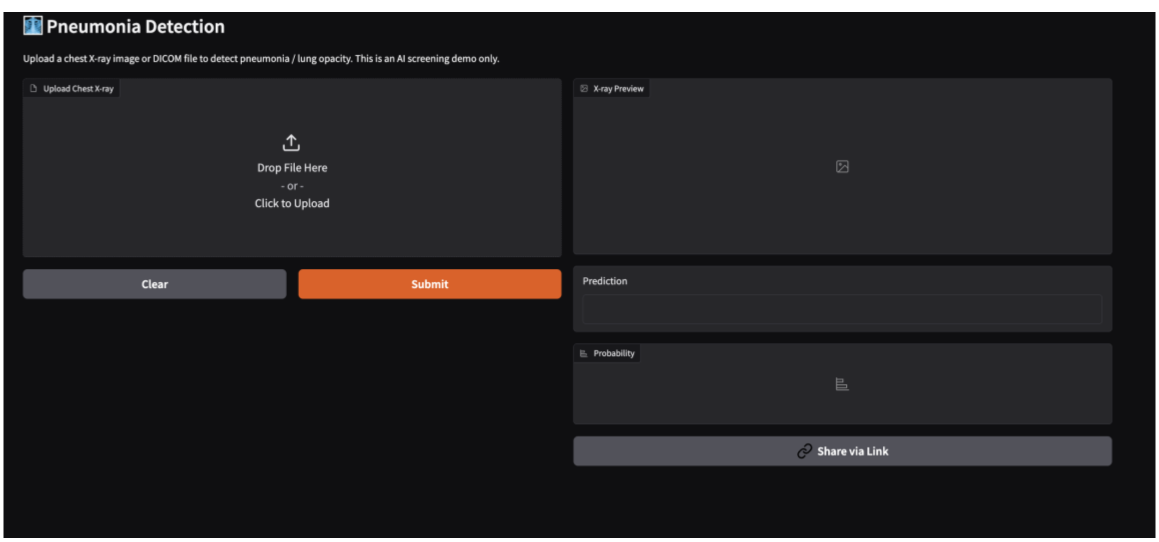
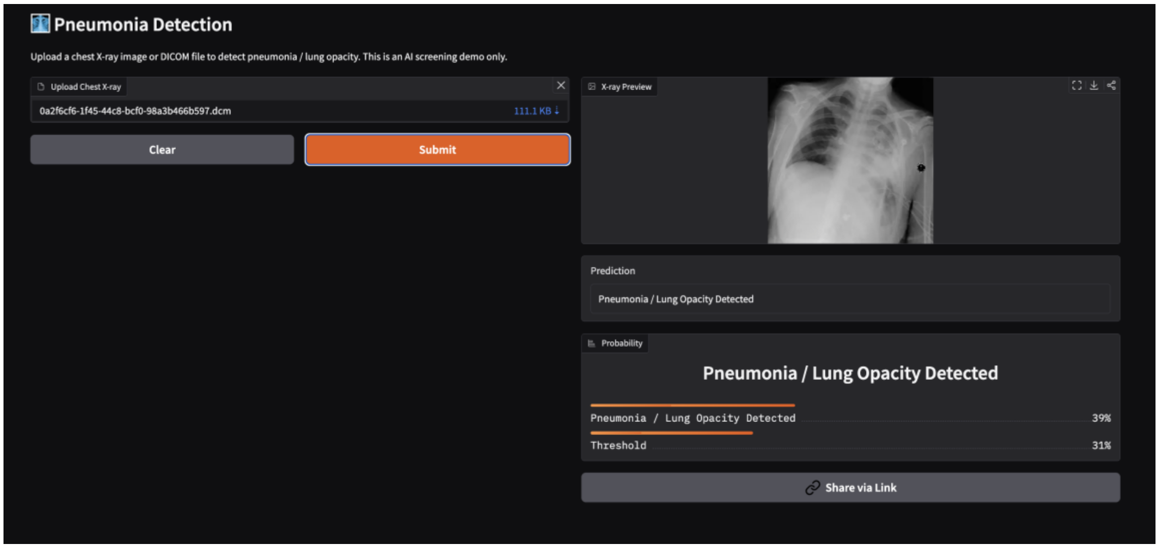


Figure 15: App for Hospital:




Pneumonia Detection
Upload a chest X-ray image or DICOM file to detect pneumonia / lung opacity. This is an AI screening demo only.
Upload Chest X-ray
0a2f6cf6-1f45-44c8-bcf0-98a3b466b597.dcm
111.1 KB
Clear
Submit
X-ray Preview
Prediction
Pneumonia / Lung Opacity Detected
Probability
Pneumonia / Lung Opacity Detected
Pneumonia / Lung Opacity Detected 39%
Threshold 31%
Share via Link